

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T24 · Solving Linear Equations

With Answers

Questions with full worked solutions

■ 28 pages

FAMILY MAP

Solving Linear Equations

F01

Clear fractional linear equations

10 Q | 33 marks

F02

Balance a linear equation

3 Q | 10 marks

Working Unit 04

4MA1-1H Q6

ORIGINAL QUESTION

3 marks

6 (a) Solve $x - 4 = \frac{3 + 2x}{6}$

Show clear algebraic working.

$x = \dots\dots\dots$
(3)

Worked Solution 04

Solve a fractional linear equation

UNIT 1
3 marks**Clear the denominator**

$$x - 4 = \frac{3 + 2x}{6} \Rightarrow 6x - 24 = 3 + 2x.$$

Collect and solve

$$4x = 27 \Rightarrow x = \frac{27}{4} = 6.75.$$

Final answer

$$x = \frac{27}{4} = 6.75$$

Question 03

4MA1-1H Q6

ORIGINAL QUESTION

3 marks

(b) Solve $3n - 4 = \frac{5n + 6}{3}$

Show clear algebraic working.

$n = \dots\dots\dots$
(3)

Worked Solution 03

Solve a linear equation

MODULAR UNIT 1

3 marks

Clear the denominator

$$3n - 4 = \frac{5n + 6}{3} \Rightarrow 9n - 12 = 5n + 6.$$

Solve

$$4n = 18 \Rightarrow n = \frac{9}{2}.$$

Final answer

$$n = \frac{9}{2}$$

Question 23

4MA1-2H Q14

ORIGINAL QUESTION

4 marks

(b) Solve $\frac{2x+3}{5} + \frac{6x-5}{4} = \frac{163}{100}$

Show clear algebraic working.

$x = \dots\dots\dots$
(4)

Worked Solution 23

Fractional equation

MODULAR UNIT 1

4 marks

Clear the denominators

$$20(2x + 3) + 25(6x - 5) = 163.$$

Solve

$$190x = 228 \Rightarrow x = 1.2.$$

Final answer

$$x = 1.2$$

Question 05

4MA1-1H Q2 (b)

ORIGINAL QUESTION UNIT 1

4 marks

There are 56 metal bars in a box.
Each metal bar is gold or silver or zinc.

y metal bars are gold.

$(3y + 7)$ metal bars are silver.

$(2y - 5)$ metal bars are zinc.

- (b) Work out the number of metal bars that are zinc.
Show clear algebraic working.

.....
(4)

Worked solution 05

Find the number of zinc bars

MODULAR UNIT 1

4 marks

(b) – Find the number of zinc bars

Use the total

$$y + (3y + 7) + (2y - 5) = 56 \Rightarrow 6y + 2 = 56.$$

Solve for y

$$6y = 54 \Rightarrow y = 9.$$

Find zinc

$$2y - 5 = 2(9) - 5 = 13.$$

Final answer

13 zinc bars

Question 16

4MA1-2H Q4 (a)

ORIGINAL QUESTION UNIT 1

3 marks

4 (a) Solve $\frac{2x+5}{6} = 2x-5$

Show clear algebraic working.

$x = \dots\dots\dots$
(3)

Worked solution 16

Solve a linear equation with a fraction

MODULAR UNIT 1

3 marks

(a) – Solve a linear equation with a fraction

Clear the denominator

$$\frac{2x + 5}{6} = 2x - 5 \Rightarrow 2x + 5 = 6(2x - 5) = 12x - 30.$$

Collect terms

$$35 = 10x.$$

Solve

$$x = \frac{35}{10} = 3.5.$$

Final answer

$$x = 3.5$$

Question 21

4MA1-2H Q15

ORIGINAL QUESTION

3 marks

(c) Solve $\frac{1-2y}{3} = \frac{4}{5} - \frac{2y-1}{2}$

Show clear algebraic working.

$y = \dots\dots\dots$
(3)

Worked Solution 21

Solve a fractional linear equation

MODULAR UNIT 1

3 marks

Clear denominators

$$30 \left(\frac{1 - 2y}{3} \right) = 30 \left(\frac{4}{5} - \frac{2y - 1}{2} \right).$$

Expand and collect

$$10 - 20y = 24 - 15(2y - 1) = 39 - 30y \Rightarrow 10y = 29.$$

Solve

$$y = \frac{29}{10} = 2.9.$$

Final answer

$$y = 2.9$$

Original question 03

4MA1-1H Q3 M03

MODULAR UNIT 1

3 marks

(c) Solve $\frac{2x+5}{3} = 4-x$

Show clear algebraic working.

$x = \dots\dots\dots$
(3)

Worked solution 03

Solve the linear equation

MODULAR UNIT 1

3 marks

M03 – Solve the linear equation**Clear the denominator**

$$(2x + 5)/3 = 4 - x \Rightarrow 2x + 5 = 12 - 3x.$$

Collect x terms

$$5x = 7.$$

Divide

$$x = 7/5.$$

Final answer

$$x = \frac{7}{5}$$

Original question 22

4MA1-2H Q12 M01

MODULAR UNIT 1

3 marks

12 (a) Solve $\frac{3x+2}{5} - \frac{2x+1}{3} = x$

Show clear algebraic working.

$x = \dots\dots\dots$
(3)

Worked solution 22

Solve the fractional equation

MODULAR UNIT 1

3 marks

M01 – Solve the fractional equation**Clear denominators**

$$3(3x + 2) - 5(2x + 1) = 15x.$$

Collect terms

$$9x + 6 - 10x - 5 = 15x \Rightarrow -x + 1 = 15x.$$

Solve

$$x = 1/16.$$

Final answer

$$x = \frac{1}{16}$$

Original question 03

4MA1-1H Q5 Q5(b)

ORIGINAL QUESTION**3 marks**

(b) Solve $2x - 3 = \frac{3x - 5}{4}$

Show clear algebraic working.

$x = \dots\dots\dots$
(3)

Worked solution 03

Chapter: Solving Linear Equations

MODULAR UNIT 1

3 marks

Q5(b) – Chapter: Solving Linear Equations

Family: Balance a linear equation • Variant: Linear unknown on both sides

MethodFrom $2x - 3 = (3x - 5)/4$, multiply by 4 to get $8x - 12 = 3x - 5$.*Why: Clearing the denominator preserves the equality.***Method** $8x - 3x = 12 - 5$, so $5x = 7$.*Why: Collect the x-terms on one side and constants on the other.***Method**Therefore $x = 7/5$.*Why: Divide both sides by 5.***Final answer****x:** $x = 7/5$

Original question 23

4MA1-2H Q15 Q15(a)

ORIGINAL QUESTION**4 marks**

15 (a) Solve $\frac{4x + 5}{3} - \frac{3 - 2x}{2} = 13$

Show clear algebraic working.

$x = \dots\dots\dots$
(4)

Worked solution 23

Chapter: Solving Linear Equations

MODULAR UNIT 1

4 marks

Q15(a) – Chapter: Solving Linear Equations

Family: Clear fractional linear equations • Variant: Several fractional linear terms

MethodMultiply by 6: $2(4x + 5) - 3(3 - 2x) = 78$.*Why: Use the lowest common denominator to clear both fractions.***Method** $8x + 10 - 9 + 6x = 78$.*Why: Expand both brackets, preserving the subtraction.***Method** $14x = 77$.*Why: Collect x -terms and constants.***Method** $x = 77/14 = 11/2 = 5.5$.*Why: Divide by 14 and simplify.***Final answer****x:** $x = 5.5$

(d) Solve $3(2x - 5) = \frac{9 - x}{2}$

Show clear algebraic working.

$x = \dots\dots\dots$
(4)

Worked solution

January 2020 | Question 7 | 4 marks

Family: Clear fractional linear equations • Variant: Single fractional linear term

Method / working

1. $3(2x-5)=(9-x)/2.$

2. Multiply by 2: $12x-30=9-x.$

3. $13x=39.$

4. $x=3.$

Final answer: $x=3$

(b) Solve $4 - 3x = \frac{5 - 8x}{4}$

Show clear algebraic working.

$x = \dots\dots\dots$
(3)

Worked solution

January 2021 | Question 5 | 3 marks

Family: Clear fractional linear equations • Variant: Single fractional linear term

Method / working

1. Multiply the printed equation $4-3x=(5-8x)/4$ by 4.

2. $16-12x=5-8x$, so $11=4x$.

3. Therefore $x=11/4=2.75$.

Final answer: $x=2.75$

- 2 Solve $3(2 - 4x) = 5 - 8x$
Show clear algebraic working.

$x = \dots\dots\dots$

(Total for Question 2 is 3 marks)

Answer reference | January 2023 | Question 2

Family: Balance a linear equation • Variant: Linear unknown on one side

2	$6-12x$ or $2-4x = \frac{5}{3} - \frac{8}{3}x$		3	M1 for expansion of bracket on the LHS or dividing the RHS by 3 with two terms
	$6-5=12x-8x$ or $1=4x$ or $-12x+8x=5-6$ oe or $-4x=-1$ or $\frac{8}{3}x-4x=\frac{5}{3}-2$ oe or $2-\frac{5}{3}=-\frac{8}{3}x+4x$ oe			M1 ft (dep on 4 terms) for terms in x on one side of equation; number terms on the other
	<i>Working required</i>	$\frac{1}{4}$		A1 oe dep on M1 awarded
				Total 3 marks

Worked solution

January 2023 | Question 2 | 3 marks

Family: Balance a linear equation • Variant: Linear unknown on one side

Official final mark-scheme pages are embedded immediately after this question.